

Who Verifies the Verifier?

Measuring LLM-as-Judge Reliability Across 32,688 Automated Research Verdicts

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32,688

claim-level verdicts

6,398

reproduction attempts

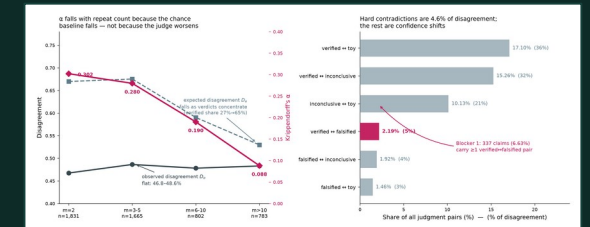
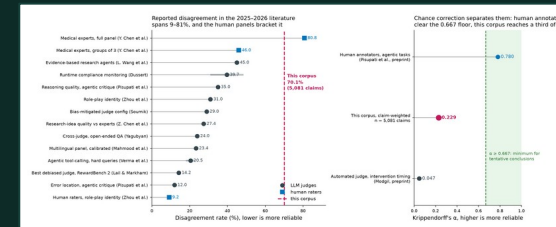
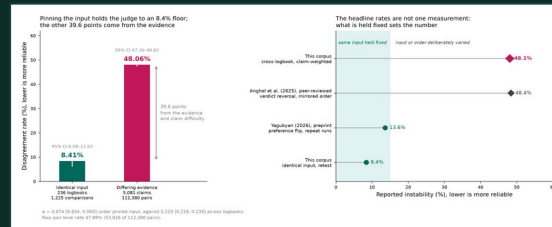
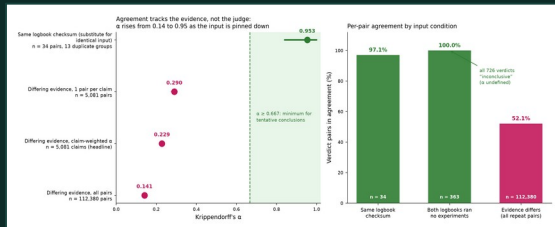
5,081

claims judged more than once

70.1%

of those carry a disagreement

A single judge model read the same research claim up to 42 times, and changed its answer on 70 percent of the claims it read twice.



Full-size figures on slides 4, 5, 6 and in the appendix

Analysis scripts, derived tables and figures: github.com/SturdivantAI/who-verifies-the-verifier

AWS-MLU Fall AI Teaching & Research Symposium
Amazon HQ2, Arlington, Virginia · September 22, 2026



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Paper & data

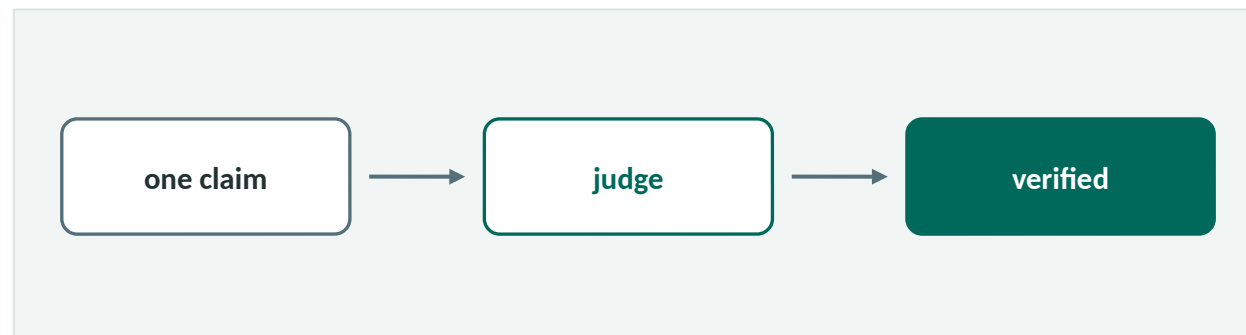
The instrument nobody validated

Large Language Models (LLMs) now score the work of other artificial intelligence (AI) systems. These judges gate reproducibility assessment, benchmark scoring, and agent evaluation.

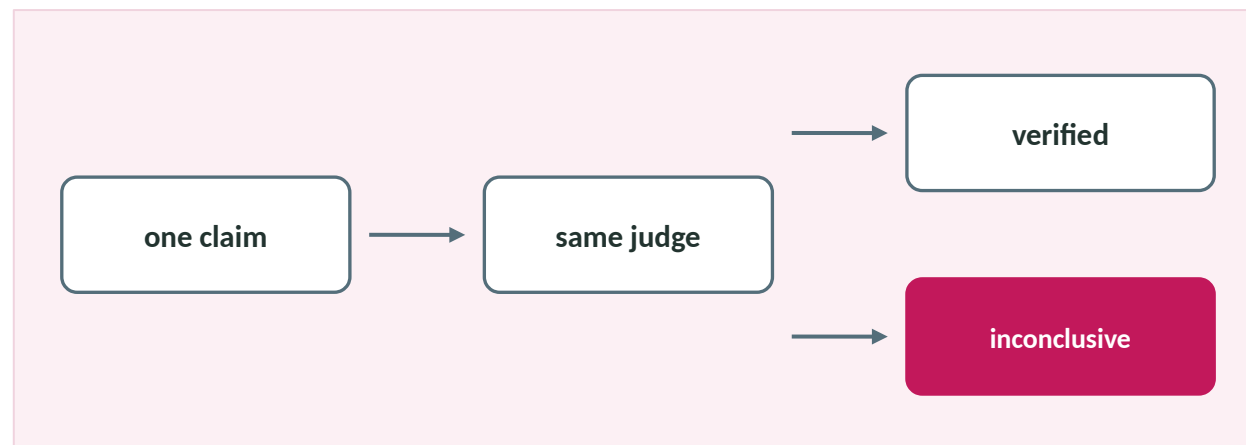
Measurement theory asks a prior question of any instrument. Does it return the same reading on the same object?

Published work reports how well judges agree with human raters. Very little of it reports whether a judge agrees with itself.

What we assume



What the log shows

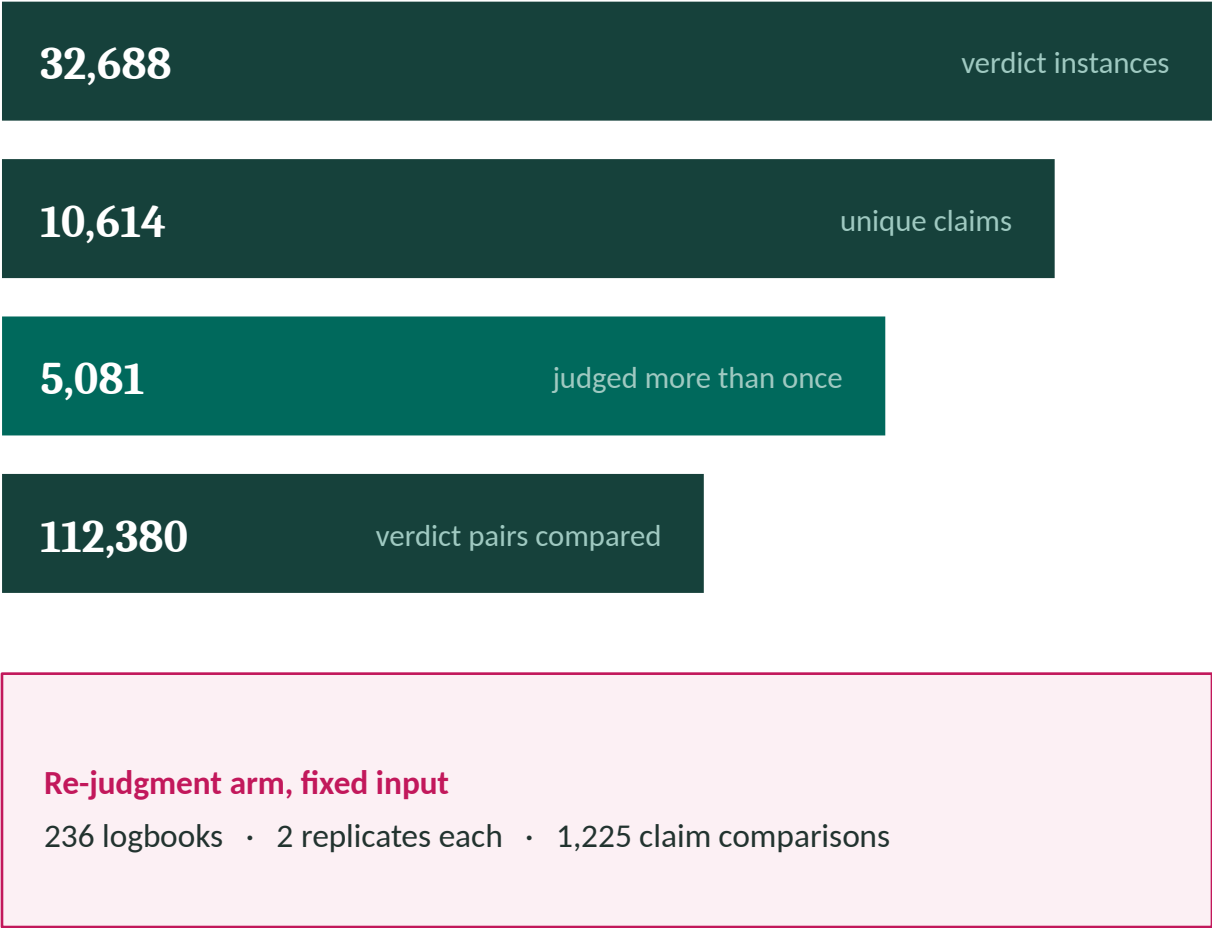


Corpus and method

The corpus is a public record of 32,688 claim-level verdicts from one judge model across 6,398 independent reproduction attempts of 2026 International Conference on Machine Learning (ICML) papers, each labelling a claim as verified, falsified, inconclusive, or toy.

Claim identity keys on paper identifier plus full claim text, isolating 10,614 unique claims. Of those, 5,081 were judged more than once, and one drew 42 separate verdicts.

A second experiment re-ran the judge on 236 logbooks, two replicates each, under a pinned model route with the system message checksum-verified. Fixing the input isolates the judge's instability.

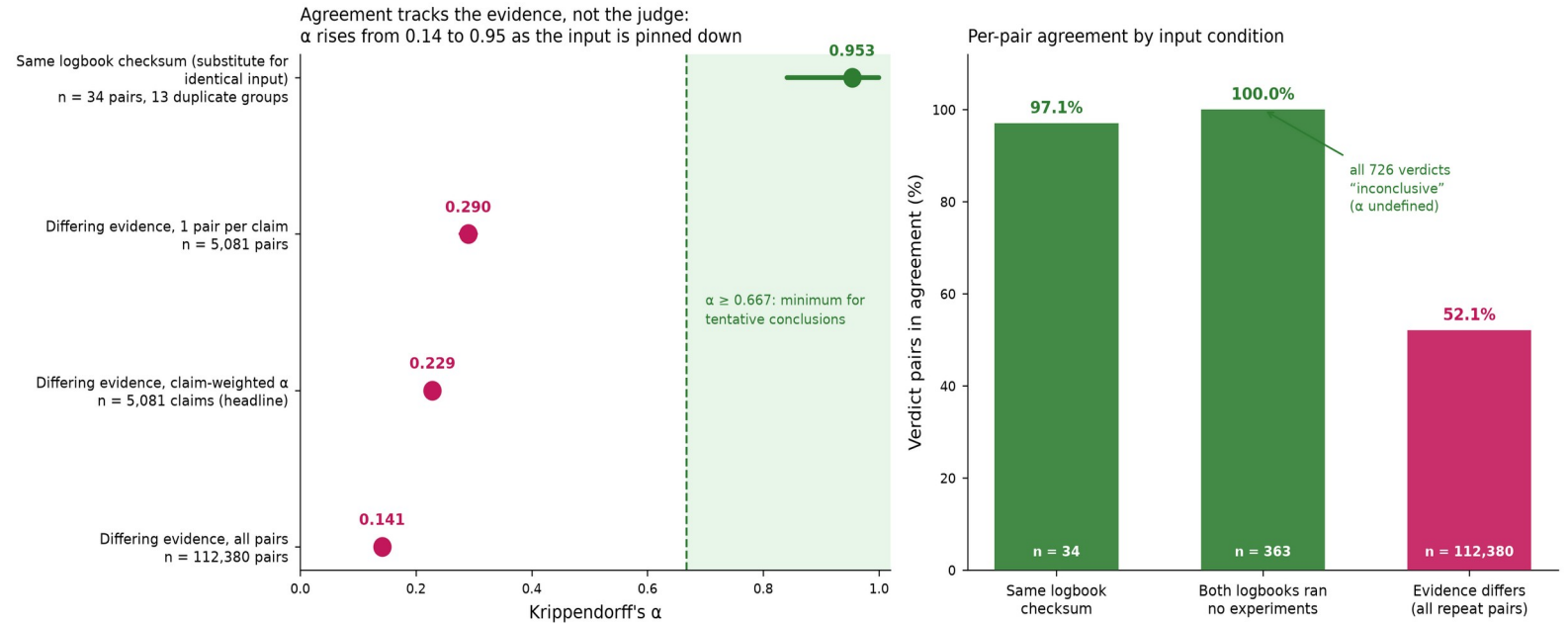


How often the judge contradicts itself

Of the 5,081 repeatedly judged claims, 3,563 carry at least one disagreeing verdict from the same judge model. That is 70.1 percent of them.

Weighting every claim equally, two readings of one claim differ 48.06 percent of the time, 95 percent confidence interval 47.30 to 48.82. Weighting every pair equally, 53,816 of 112,380 pairs disagree, or 47.89 percent.

Claim-weighted Krippendorff's alpha across the four labels is 0.229, interval 0.218 to 0.239. Krippendorff's floor for tentative conclusions is 0.667.



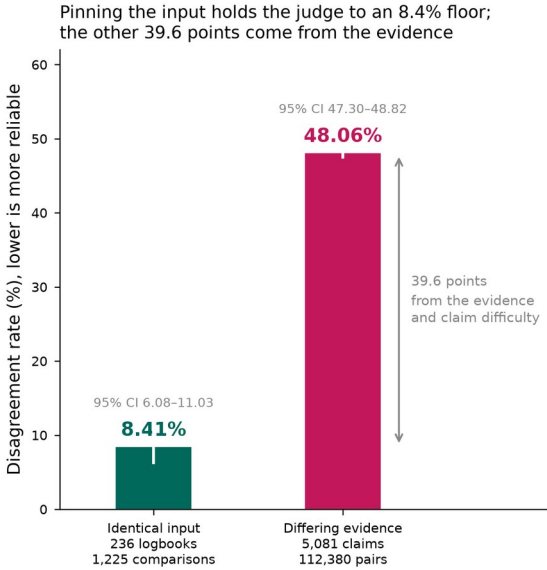
Left: four alpha values under four stated conditions, against the 0.667 floor. Right: per-pair agreement by input condition.

Is it the judge, or the evidence?

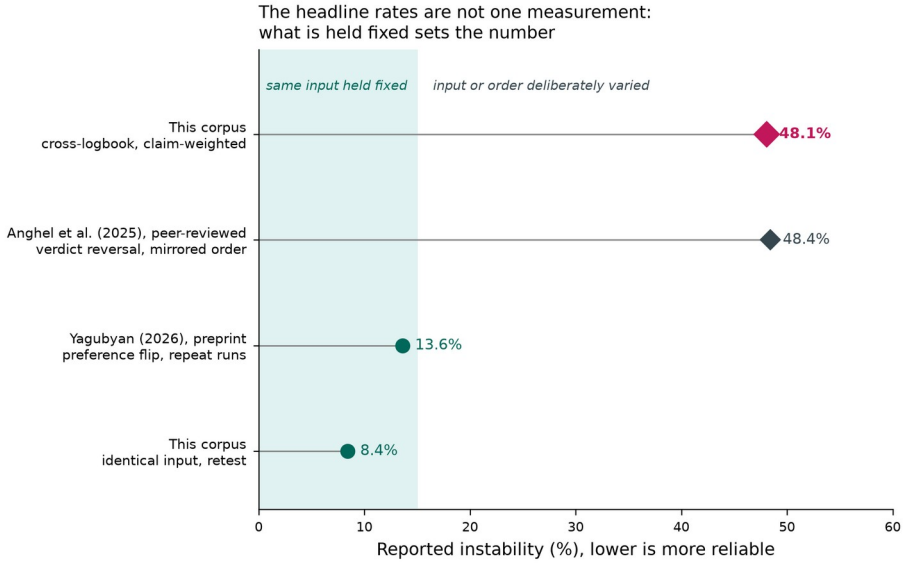
Disagreement on its own does not identify its source. Two readings of one claim can differ because the judge is unstable, or because two reproduction attempts placed different evidence in front of it.

Fixing the evidence answers it. Across 236 logbooks and 1,225 claim comparisons on byte-identical input, the judge disagreed with itself 8.41 percent of the time, interval 6.08 to 11.03, with alpha of 0.874. Of 236 logbooks, 183 were fully stable.

That floor sits 39.6 points below the claim-weighted cross-logbook rate of 48.06 percent.



$\alpha = 0.874$ [0.834, 0.909] under pinned input, against 0.229 [0.218, 0.239] across logbooks.
Raw pair-level rate 47.89% (53,816 of 112,380 pairs).

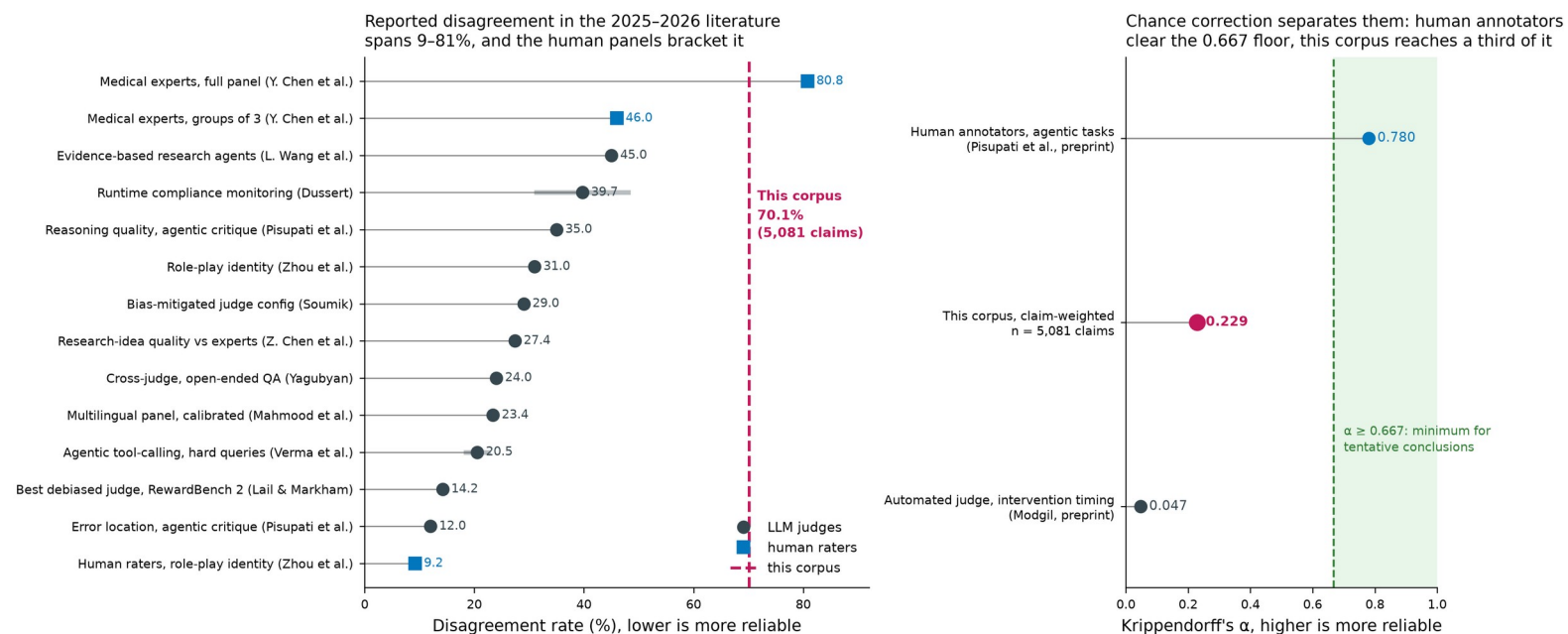


Left: the pinned-input floor against the cross-logbook rate, with the 39.6-point gap. Right: what each reported rate holds fixed.

Against the human ceiling

A disagreement rate carries little meaning without a comparator. Chen and colleagues, in JMIR AI, report average agreement of 19.2 percent among medical experts scoring summarisation quality, rising to 54 percent among groups of three. Human disagreement near 80.8 percent exceeds the 70.1 percent of claims carrying a disagreement here.

Chance-corrected figures separate the two more cleanly. Pisupati and colleagues report human inter-annotator alpha of 0.78, above the 0.667 floor. Modgil reports 0.047 for an automated judge. This corpus falls between them at 0.229, claim-weighted.



Left: reported disagreement across the 2025 to 2026 literature, human panels in blue. Right: chance-corrected alpha against the 0.667 floor.

Why this matters for AI coursework

Institutions are adopting automated evaluation for research assessment and grading, and some now build reproducibility instruction on it. This work supplies measured evidence that such instruments require validation before they are trusted, and a reusable method for measuring it.

What matters is the reliability of the instrument measuring a model's performance, not whether the model performs well.

Two practices follow. Any automated judge used for assessment should report its own test-retest rate on fixed input. Any score from one judge reading should carry the interval that rate implies.

Precedent: Software as a Medical Device (SaMD) Verification and Validation (V&V) frameworks require demonstrated reliability of the method itself, in addition to evidence about the device under test.

Reproducibility score

verified

as issued today

Reproducibility score

what it should report

verified $\pm$ one label

Judge test-retest on fixed input: 8.41%

Cross-evidence disagreement: 48.06%, claim-weighted

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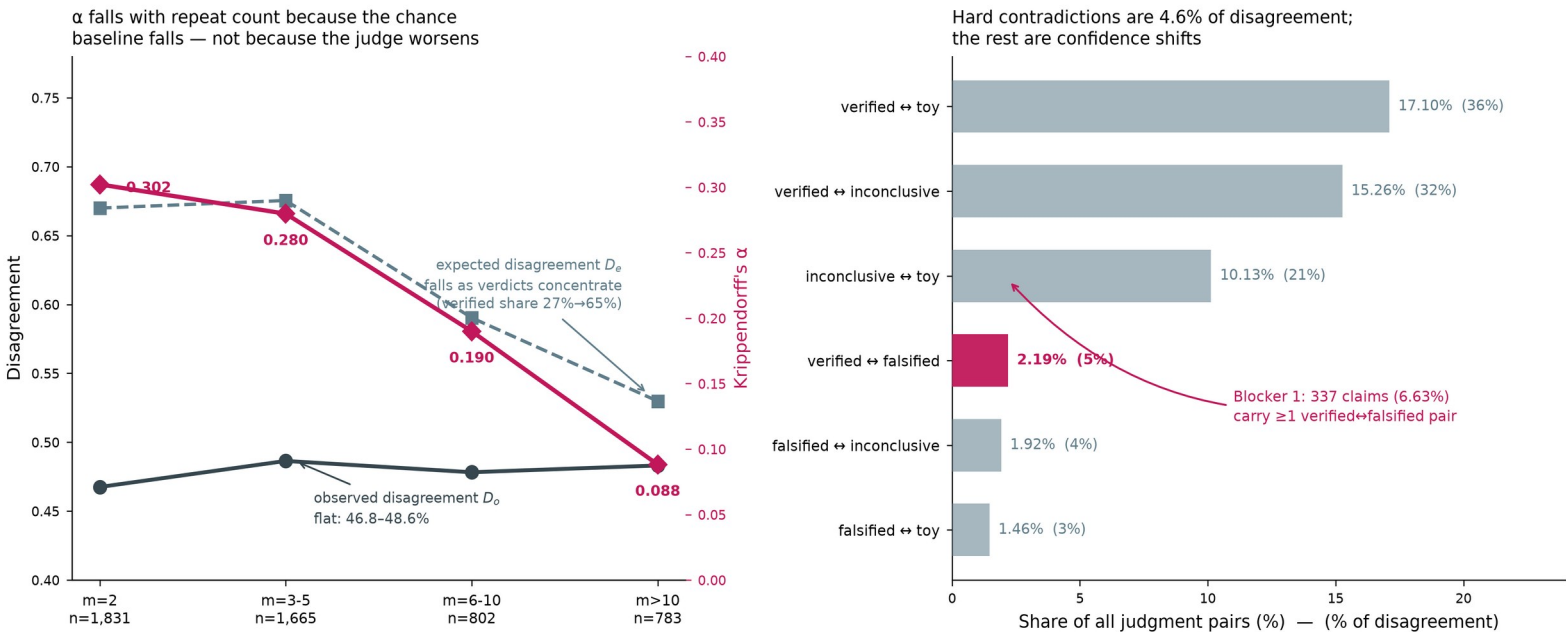
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Appendix: where the disagreement sits

Reached on question only, not part of the walkthrough.

Alpha falls as a claim is judged more often, from 0.302 at two readings to 0.088 above ten. The judge does not worsen. Observed disagreement stays flat between 46.8 and 48.6 percent while the chance baseline drops, because verdicts concentrate on verified.

Most disagreement is a confidence shift, not a reversal. Claim-weighted, verified against falsified runs at 2.19 percent of all pairs, which is 4.6 percent of disagreement. That touches 337 claims, or 6.63 percent of the 5,081.



Left: alpha by repeat-count band, with observed against expected disagreement. Right: the six label-pair transitions by share.